

Chapter 11

Question 11



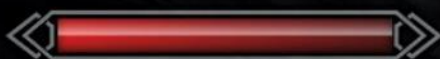
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What is Knapsack?

Recall that in the Knapsack Problem, we have n items, each with a weight w_i and a value v_i . We also have a weight bound W , and the problem is to select a set of items S of highest possible value subject to the condition that the total weight does not exceed W —that is, $\sum_{i \in S} w_i \leq W$. Here's one way to look at the approximation algorithm that we designed in this chapter. If we are told there exists a subset $\mathcal{O}$ whose total weight is $\sum_{i \in \mathcal{O}} w_i \leq W$ and whose total value is $\sum_{i \in \mathcal{O}} v_i = V$ for some V , then our approximation algorithm can find a set $\mathcal{A}$ with total weight $\sum_{i \in \mathcal{A}} w_i \leq W$ and total value at least $\sum_{i \in \mathcal{A}} v_i \geq V/(1 + \epsilon)$. Thus the algorithm approximates the best value, while keeping the weights strictly under W . (Of course, returning the set $\mathcal{O}$ is always a valid solution, but since the problem is NP-hard, we don't expect to always be able to find $\mathcal{O}$ itself; the approximation bound of $1 + \epsilon$ means that other sets $\mathcal{A}$, with slightly less value, can be valid answers as well.)



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Our Question

Now, as is well known, you can always pack a little bit more for a trip just by “sitting on your suitcase”—in other words, by slightly overflowing the allowed weight limit. This too suggests a way of formalizing the approximation question for the Knapsack Problem, but it's the following, different, formalization.

Suppose, as before, that you're given n items with weights and values, as well as parameters W and V ; and you're told that there is a subset $\mathcal{O}$ whose total weight is $\sum_{i \in \mathcal{O}} w_i \leq W$ and whose total value is $\sum_{i \in \mathcal{O}} v_i = V$ for some V . For a given fixed $\epsilon > 0$, design a polynomial-time algorithm that finds a subset of items $\mathcal{A}$ such that $\sum_{i \in \mathcal{A}} w_i \leq (1 + \epsilon)W$ and $\sum_{i \in \mathcal{A}} v_i \geq V$. In other words, you want $\mathcal{A}$ to achieve at least as high a total value as the given bound V , but you're allowed to exceed the weight limit W by a factor of $1 + \epsilon$.



Greedy Solution in Lesson

Round up all values:

- $0 < \varepsilon \leq 1$ = precision parameter.
- $v_{\max}$ = largest value in original instance.
- θ = scaling factor = $\varepsilon v_{\max} / 2n$.

$$\bar{v}_i = \left\lceil \frac{v_i}{\theta} \right\rceil \theta, \quad \hat{v}_i = \left\lfloor \frac{v_i}{\theta} \right\rfloor \theta$$

item	value	weight
1	934221	1
2	5956342	2
3	17810013	5
4	21217800	6
5	27343199	7

original instance ($W = 11$)

item	value	weight
1	1	1
2	6	2
3	18	5
4	22	6
5	28	7

rounded instance ($W = 11$)

Observation. Optimal solutions to problem with $\bar{v}$ are equivalent to optimal solutions to problem with $\hat{v}$.

Intuition. $\bar{v}$ close to v so optimal solution using $\bar{v}$ is nearly optimal; $\hat{v}$ small and integral so dynamic programming algorithm II is fast.

Our Questions Solution

Round up all values:

- $0 < \epsilon \leq 1$ = precision parameter.
- $w_{\max}$ = largest weight in original instance.
- θ = scaling factor = $n / (\epsilon w_{\max})$

max

$$\bar{w}_i = \left\lceil \frac{w_i}{\theta} \right\rceil$$

$$\hat{w}_i = \left\lfloor \frac{w_i}{\theta} \right\rfloor$$

Note: Do not round down to 0.

Example

Example. Suppose you're given four items, with weights and values as follows:

$$(w_1, v_1) = (5, 3), (w_2, v_2) = (4, 6)$$

$$(w_3, v_3) = (1, 4), (w_4, v_4) = (6, 11)$$

You're also given $W = 10$ and $V = 13$ (since, indeed, the subset consisting of the first three items has total weight at most 10 and has value 13). Finally, you're given $\epsilon = .1$. This means you need to find (via your approximation algorithm) a subset of weight at most $(1 + .1) * 10 = 11$ and value at least 13. One valid solution would be the subset consisting of the first and fourth items, with value $14 \geq 13$. (Note that this is a case where you're able to achieve a value strictly greater than V , since you're allowed to slightly overfill the knapsack.)

V	W	$\overline{w}_i$
3	5	4
6	4	4
4	1	1
11	6	4